

Component	Coverage	Binary FPR	Binary FNR	Note
Ensemble (RF+AdaBoost+Bagging)	All attack types, all protocols	0.0123	0.0027	Known attacks, hard voting, SHAP explainability
G-KDE Anomaly Detector	Any deviation from normal distribution	0.0469	0.0125	Zero-day capable; trained on benign only
LSTM Sequence Classifier	MQTT temporal attack patterns	0	0.0001	Temporal attacks: slow brute force, session hijacking
Combined OR-Logic System	All of the above simultaneously	0.0022	0.0022	Flag from any single component escalates session